

# USING 211 MAINE DATA FOR SYSTEM PLANNING IN MAINE

**11/13/2025**

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# OUR PROJECT

We had powerful 211 data on requests for services and community-based providers in Maine.

How could we leverage that data to improve community-based service planning?

# OUR DATA

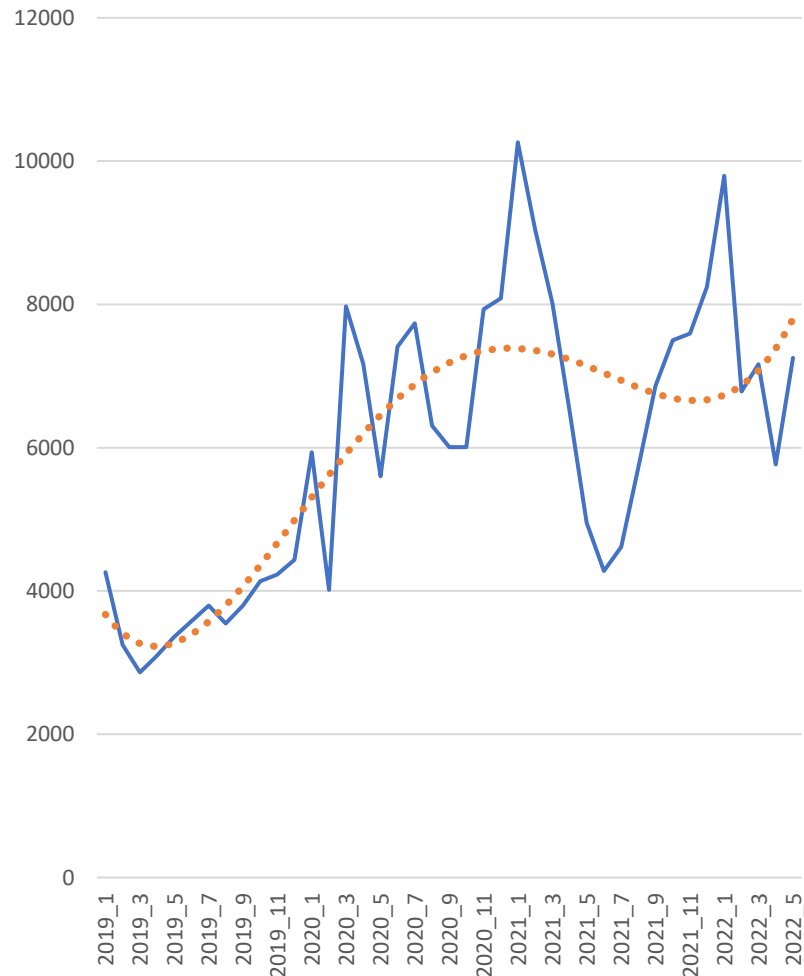
We used 211 Maine service request data as a proxy for a primary prevention population and their needs.

We used 2019 US Census American Community Survey data to describe community characteristics

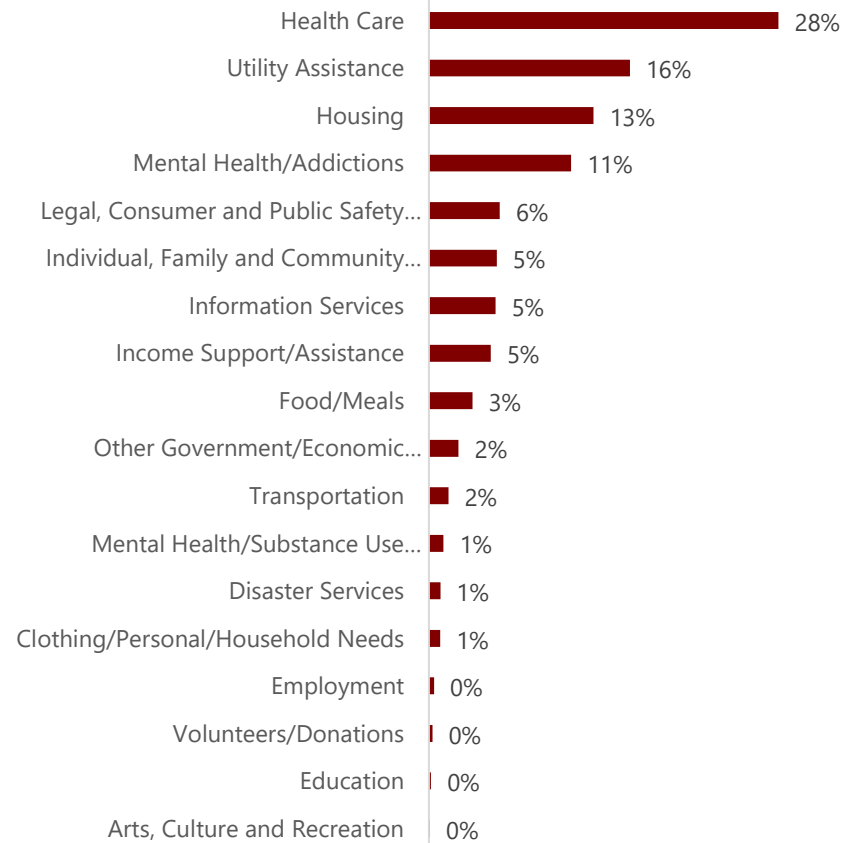
We analyzed the data Zip Code Level.

# 211 MAINE REQUEST DATA

Total 211 Requests  
January 2019 to May 2022

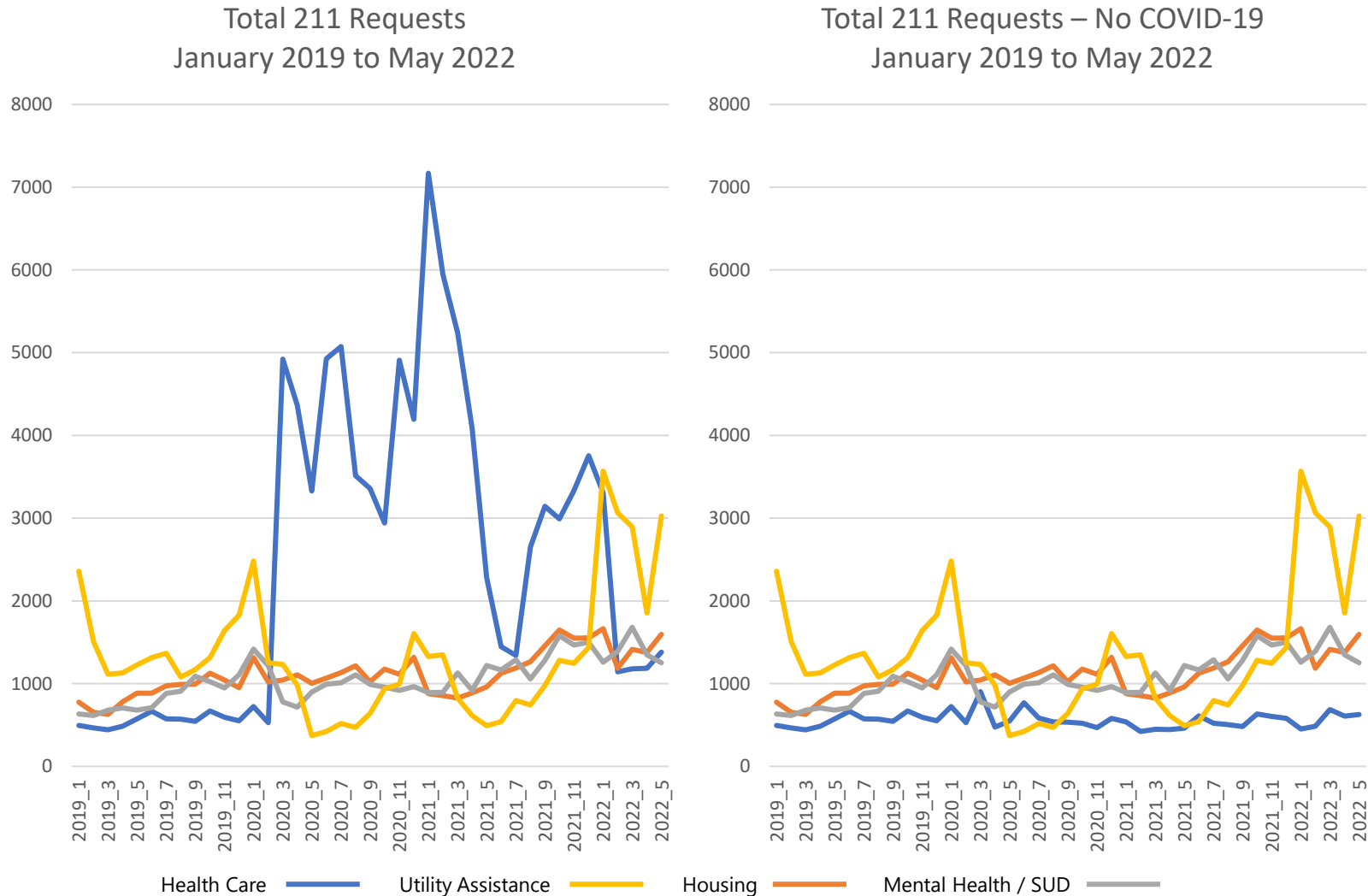


Total 211 Requests by Category  
January 2019 to May 2022



- Calls increased drastically during COVID...
- ...and never really got back to their baseline
- Calls have *seasonal* patterns
- Health care dominates people's needs
- Housing, Utilities and MH/Addictions form an important second tier

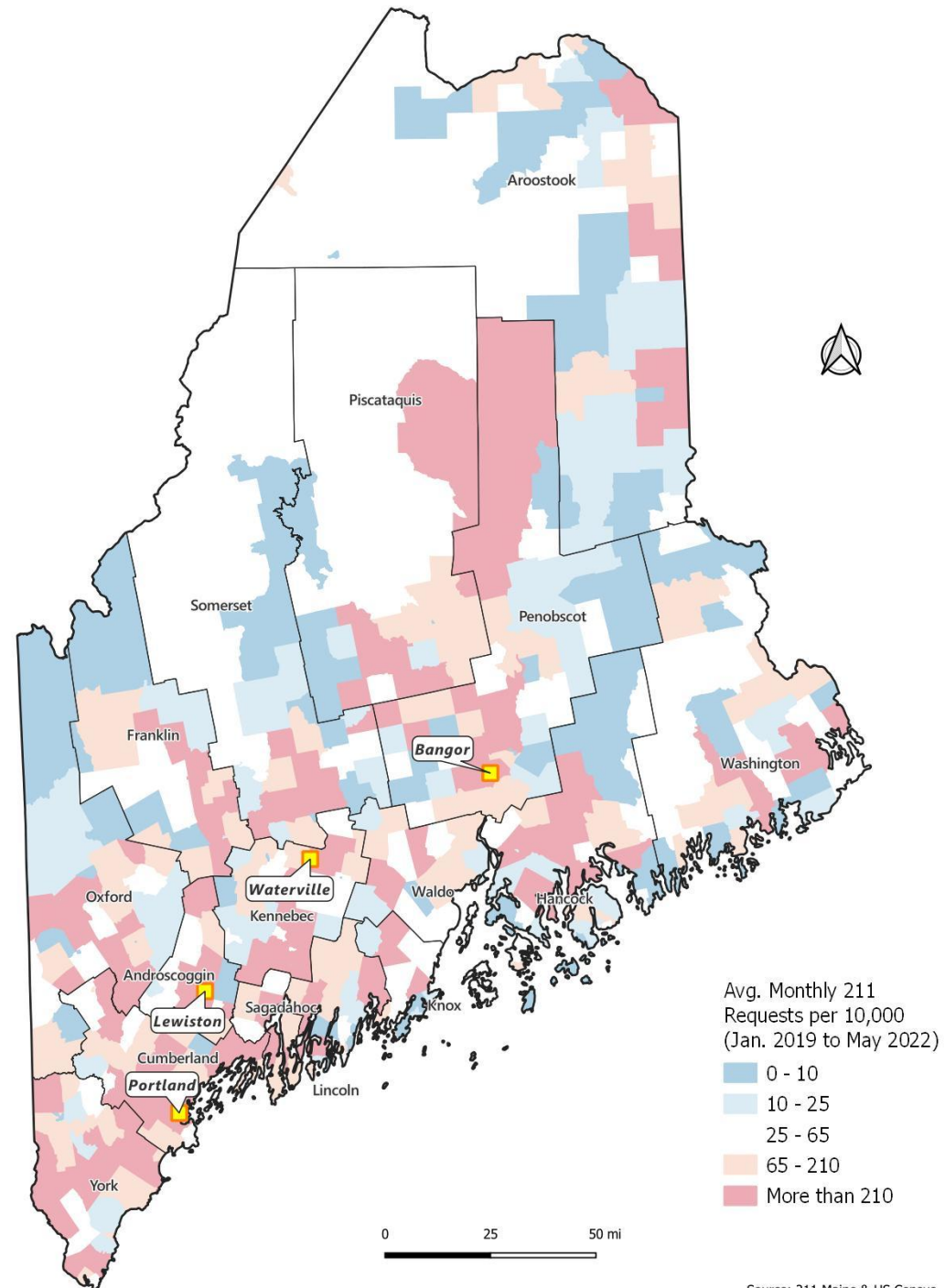
# 211 MAINE REQUEST DATA



- The impact of COVID-19 is significant in the data.
- Maine promoted 211 as the front door to their COVID-19 response.
- Early analysis indicated there was little difference in the models when the outcome was all requests or a specific type of request.

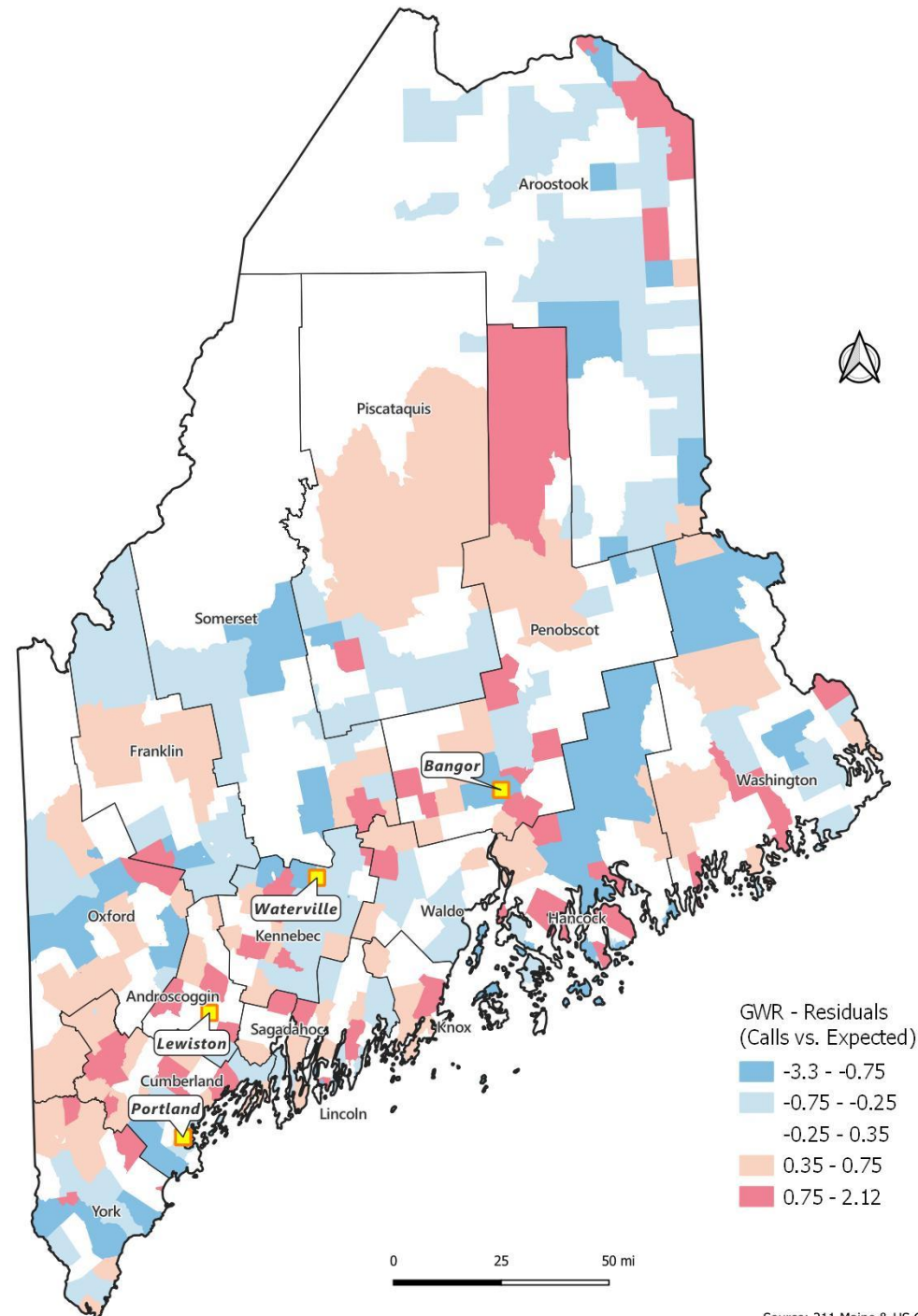
# CALLS ARE UNEVENLY DISTRIBUTED

211 calls are highly concentrated across space. They are also unevenly distributed across populations.



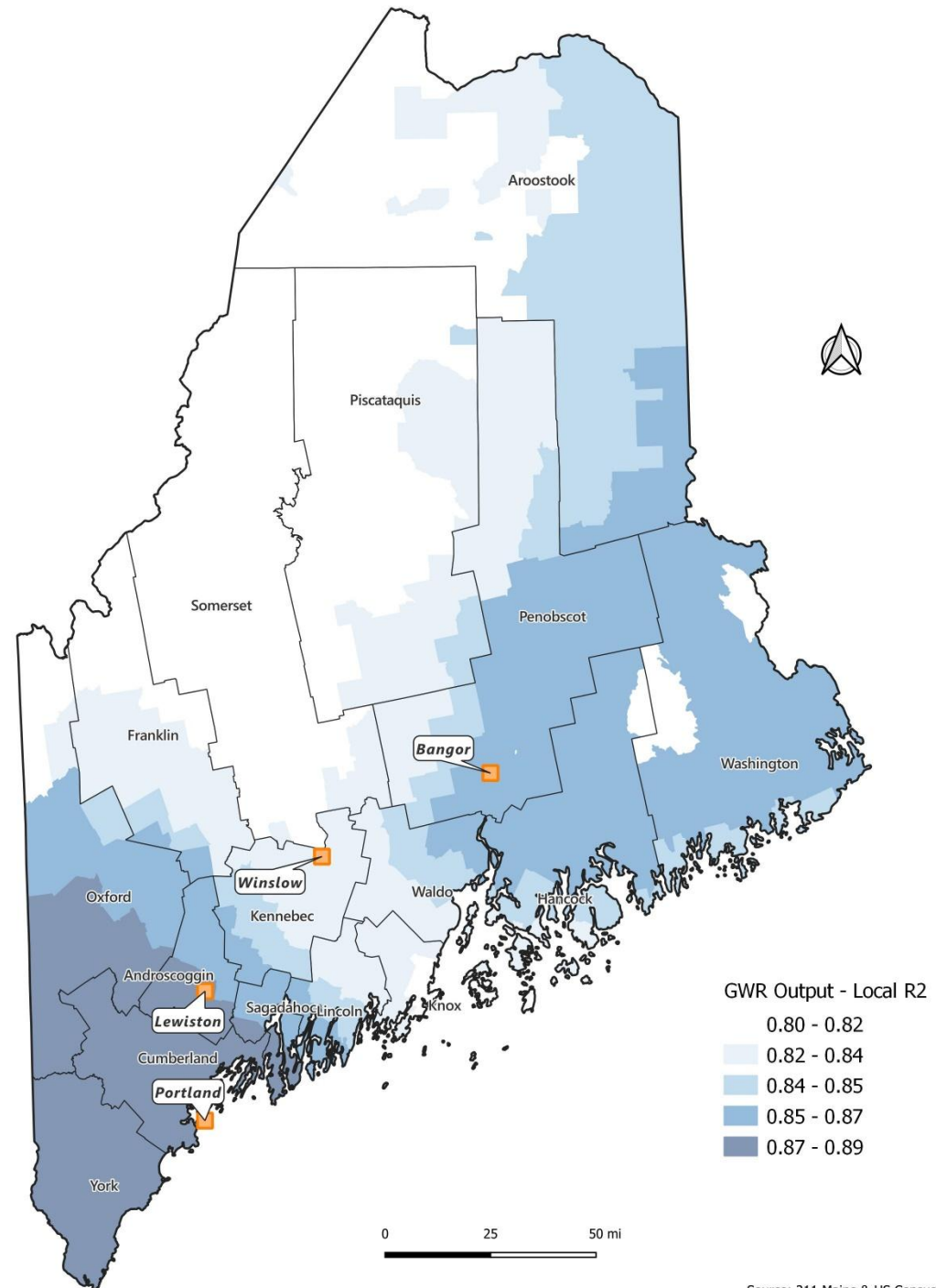
# SOME COMMUNITIES USE 211 MORE THAN OTHERS

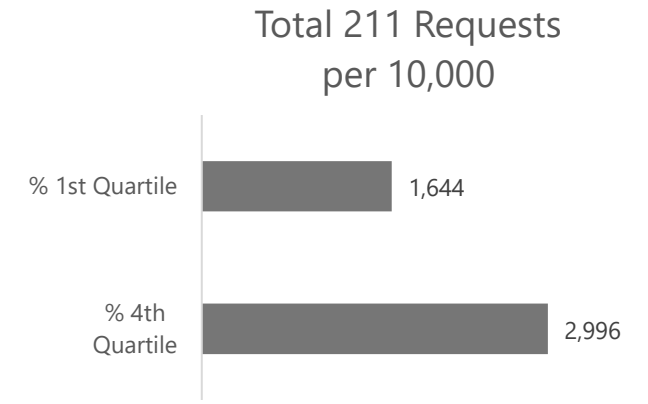
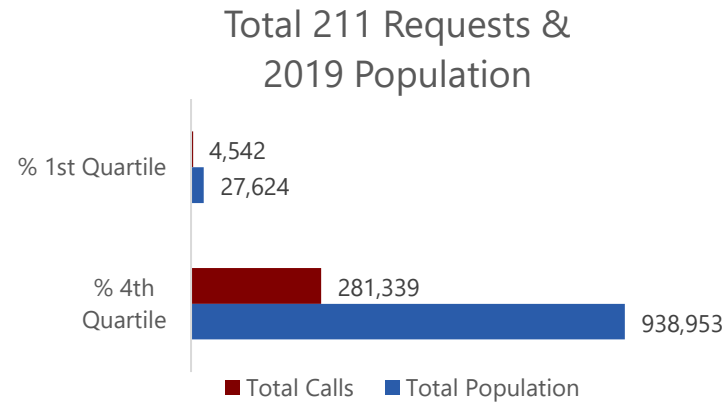
Some communities (shaded red) call 211 more than what the profile of their community would suggest. Others (shaded blue) call less than their community characteristics would suggest.



# SOME COMMUNITIES USE 211 MORE THAN OTHERS

Some communities' 211 calls are more predictable than others. The darker a region's colors, the more effectively we can predict their 211 calling patterns.

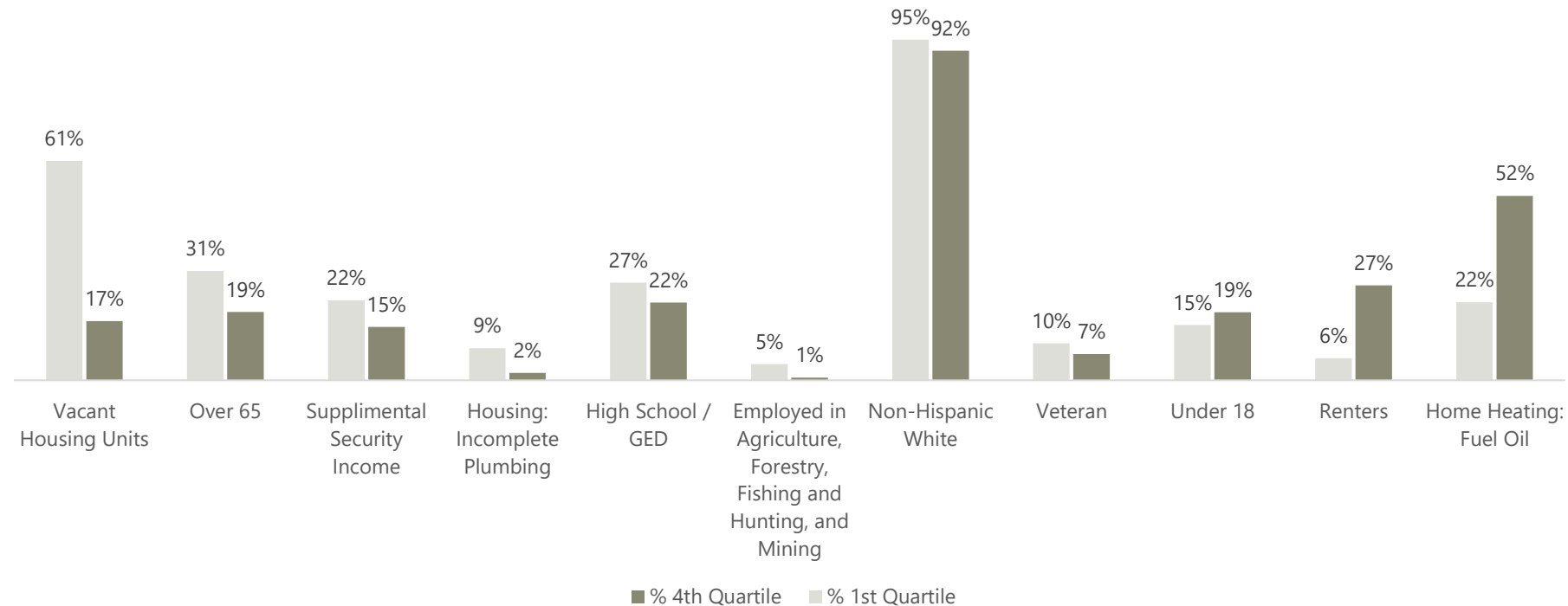




## HIGH / LOW CALL ZIP CODES ARE DIFFERENT

When comparing the zip codes that use 211 the most to those that use it the least there are significant differences in their community characteristics.

Community Characteristics by 211 Request Quartile



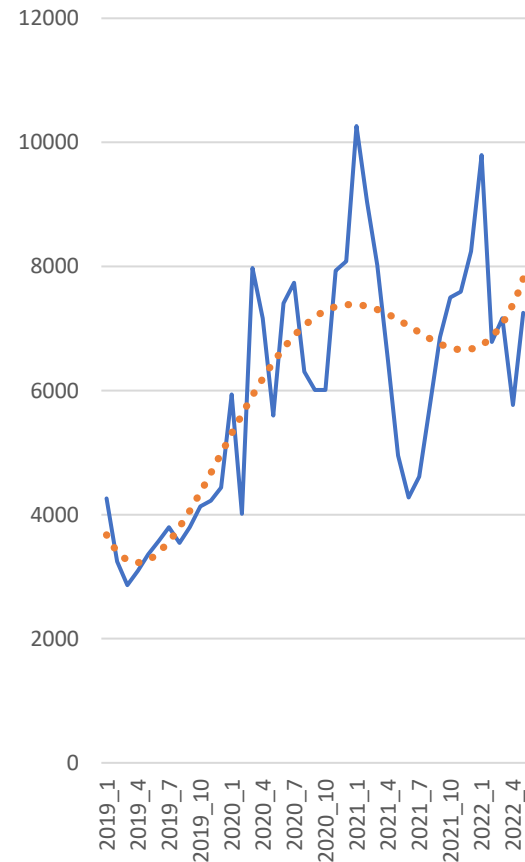
# SPATIAL GAP ANALYSIS

Identifying gaps in Maine's mental health  
and substance use disorder service array

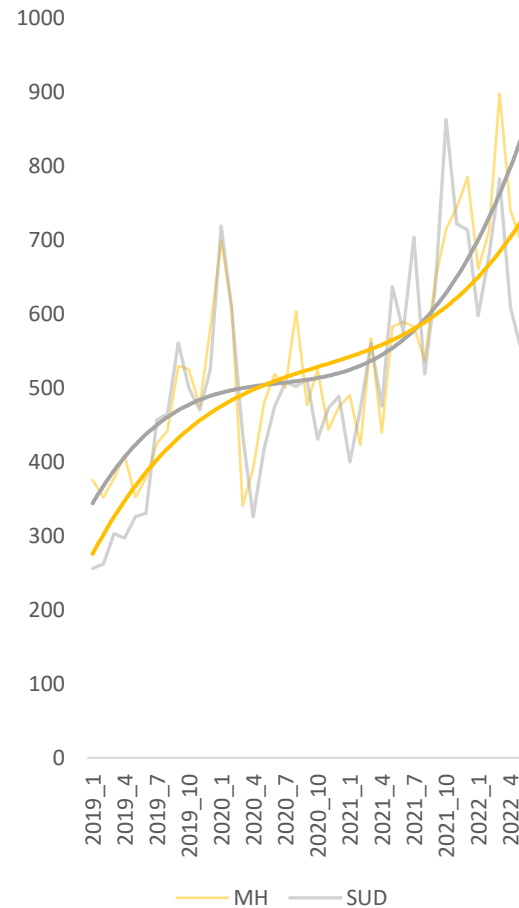
# 211 REQUEST DATA

## Mental Health & Substance Use Disorder Calls have been increasing

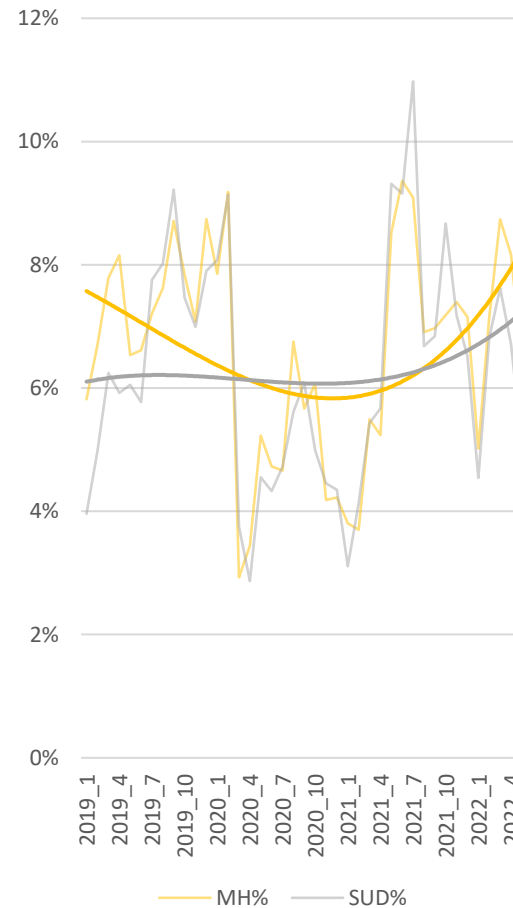
Total 211 Requests  
January 2019 to May 2022



Total MH & SUD Calls  
January 2019 to May 2022



% of Calls that are MH & SUD  
January 2019 to May 2022



## Spatial gap analysis:

A spatial gap analysis identifies imbalances between the needs of a population and the resources or services available to meet those needs.

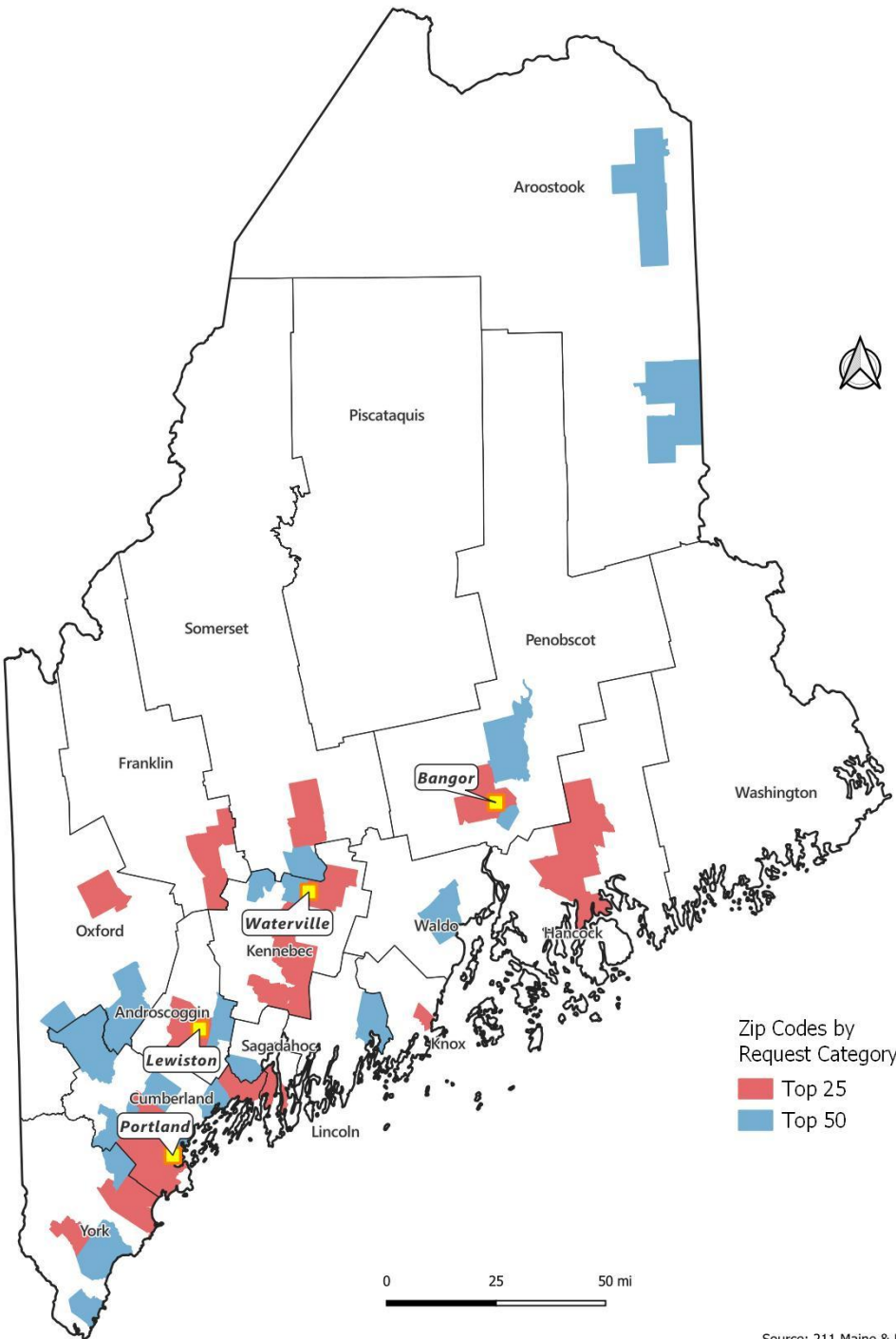
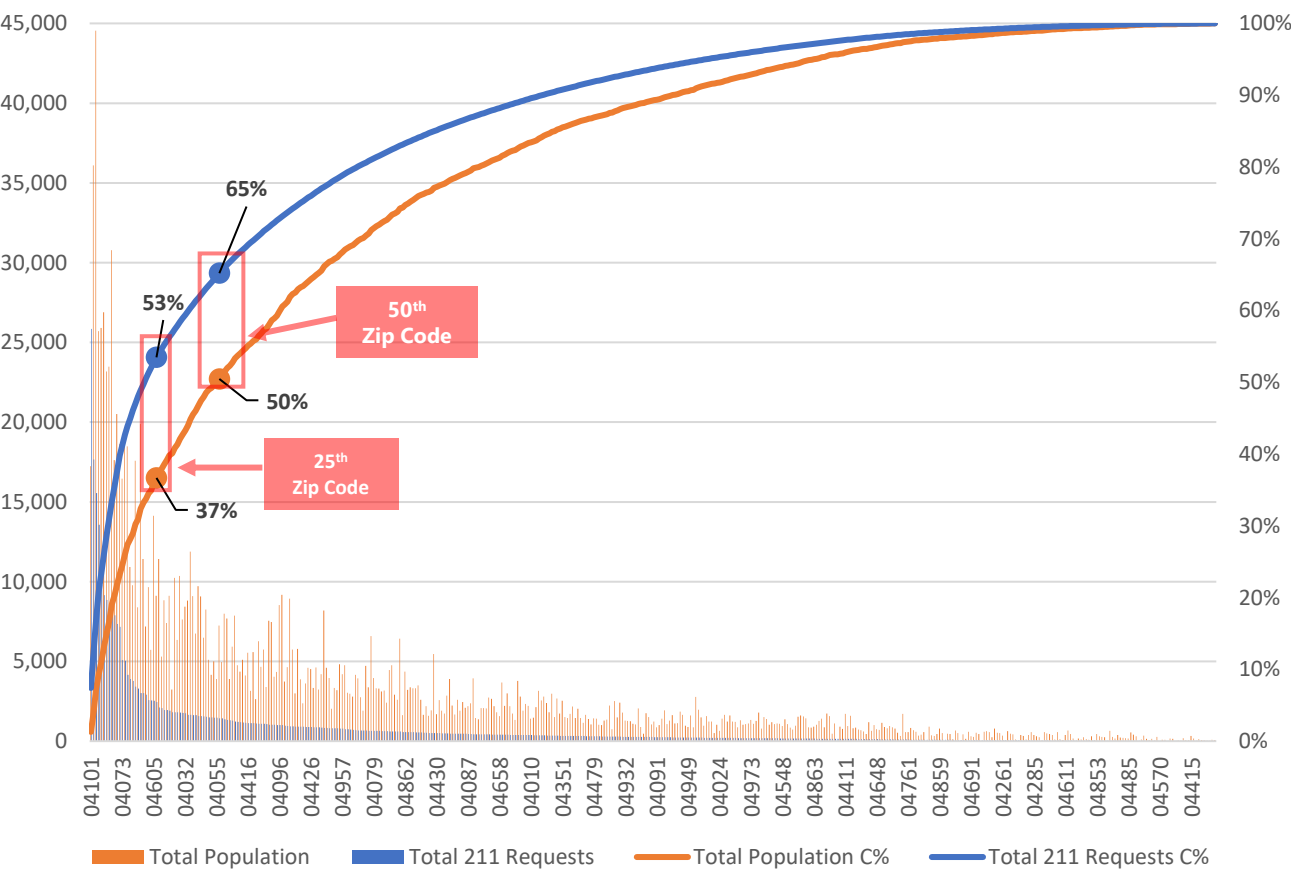
### Requires:

- Data on resources (supply) – 211 Providers
- Data on assessed needs (demand) – 211 Requests
- 214,643 calls requesting 351,913 services.
- At any given time between 6% & 8% of calls are MH or SUD related

# 211 REQUEST DATA

211 requests are highly spatially concentrated

Total 211 Requests & Total Population by Zip Code



Source: 211 Maine & US Census

# GAPS IN SERVICES

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What are Proximity Scores?

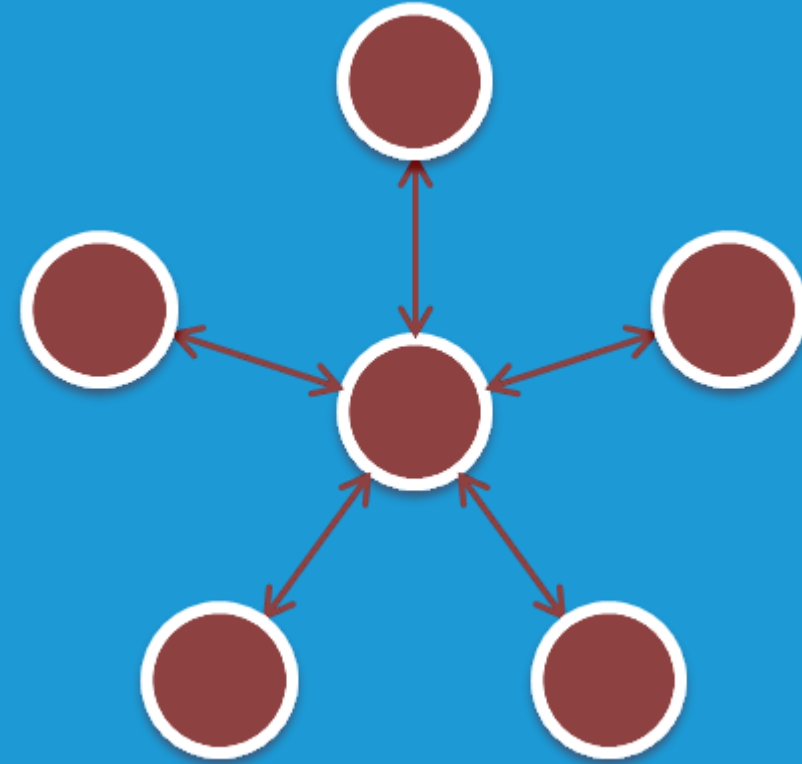
Distance to nearest provide gives a limited picture of access

Proximity scores add contextual features that account for:

1. Distance to providers
2. Availability of providers
3. Difficulty reaching providers

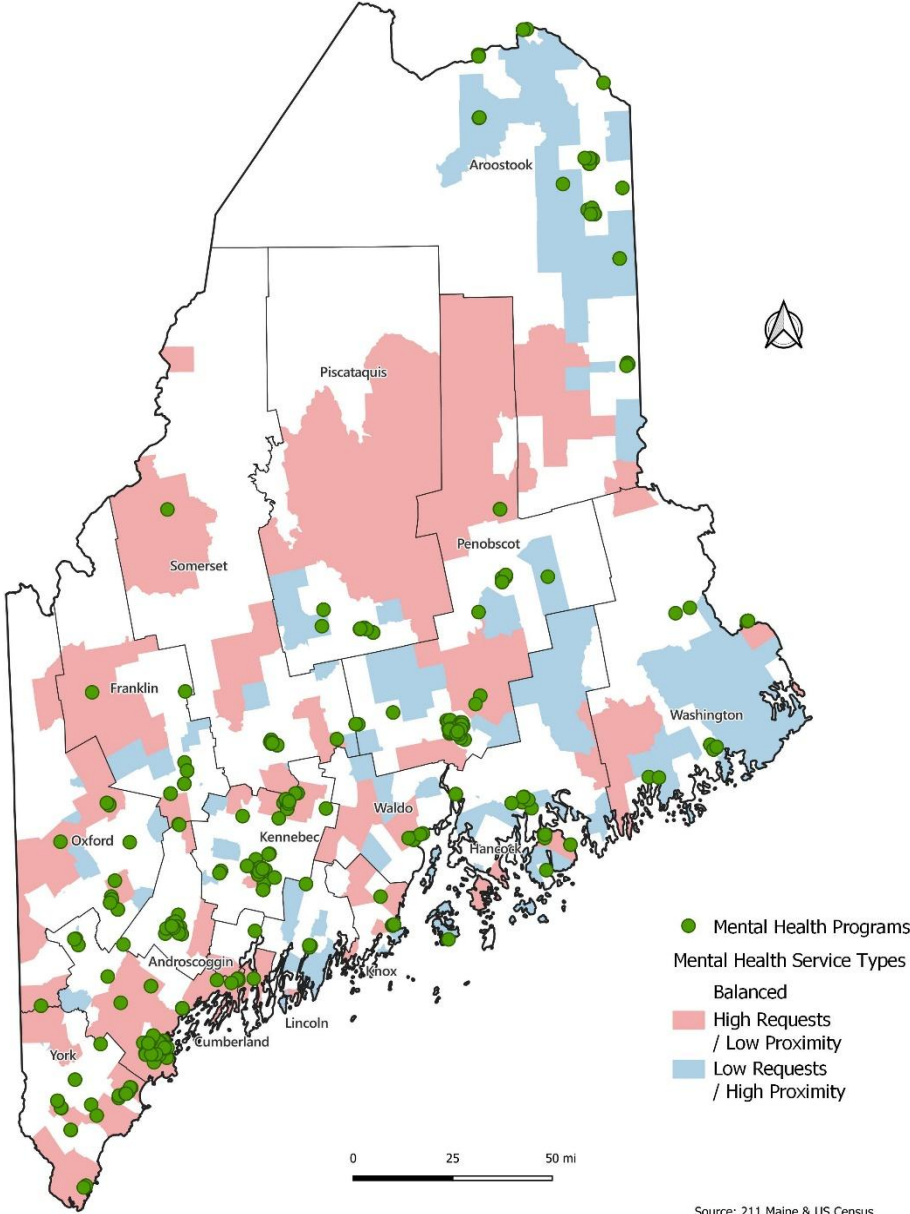
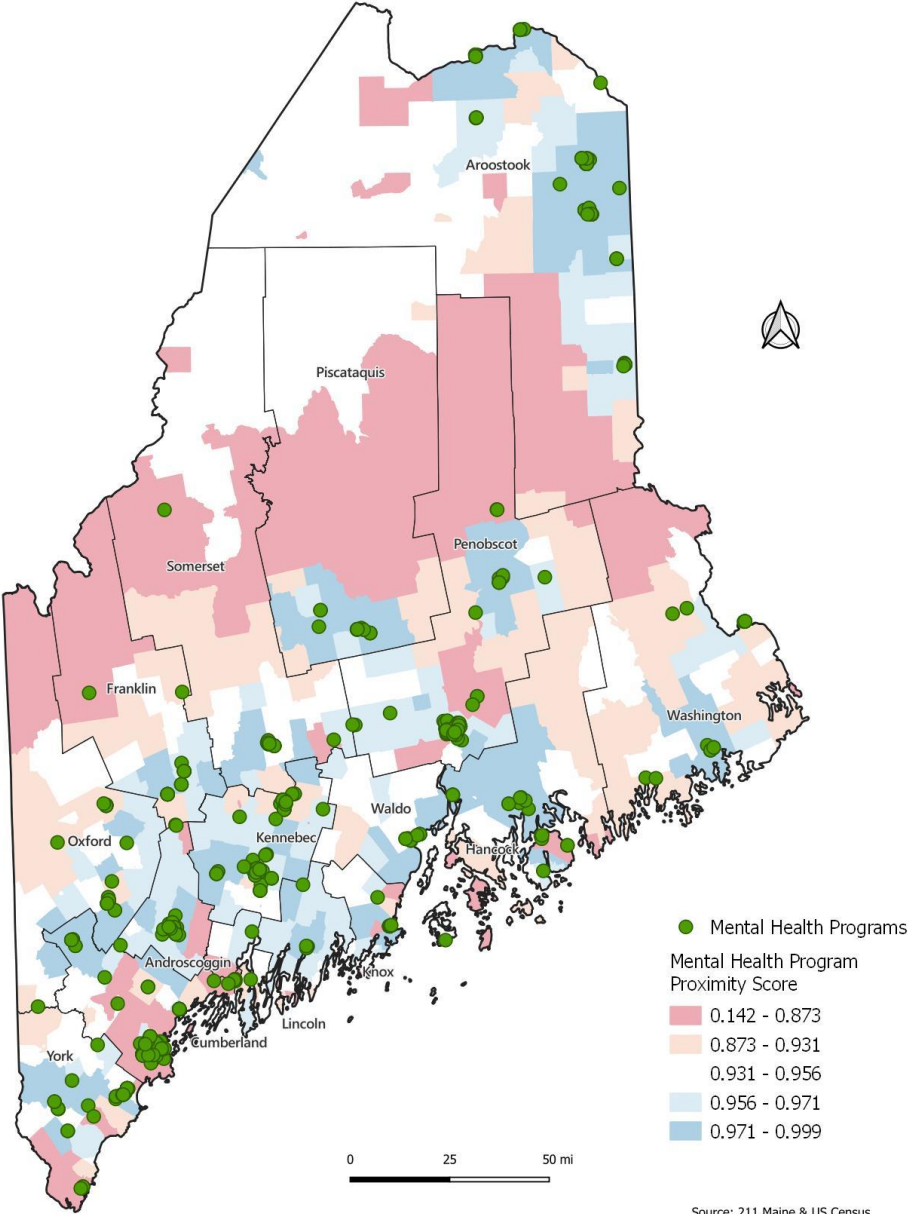
Proximity scores measure:

1. The distance between a location and the five closest service providers
2. Distance is the function by which the value of a provider is discounted - *The greater the distance the less they 'count'*
3. The effect of distance upon the value of a provider changes depending upon context - *The more urban the setting and higher the population density the more distance counts*



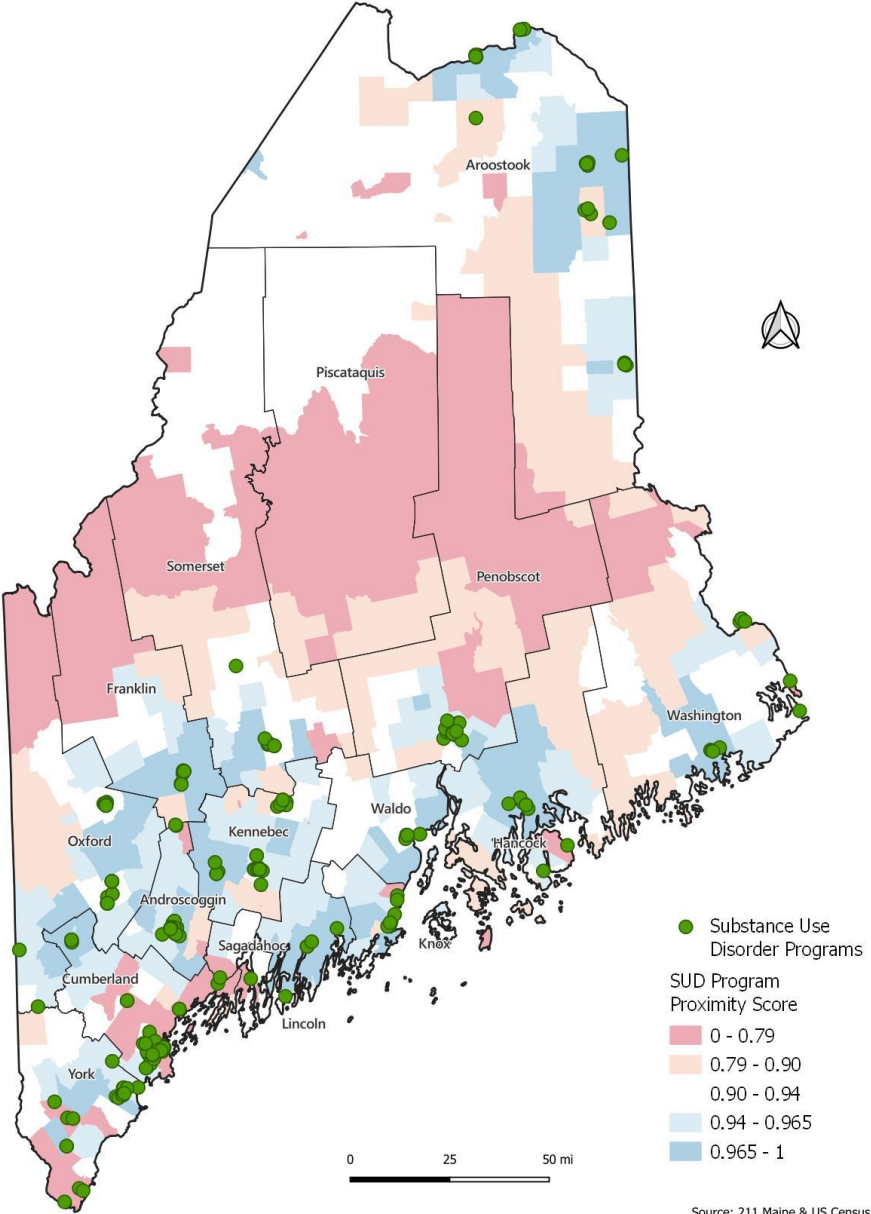
# MENTAL HEALTH SERVICES

We were able to identify 566 Mental Health programs in 211 Maine, 8% of the total

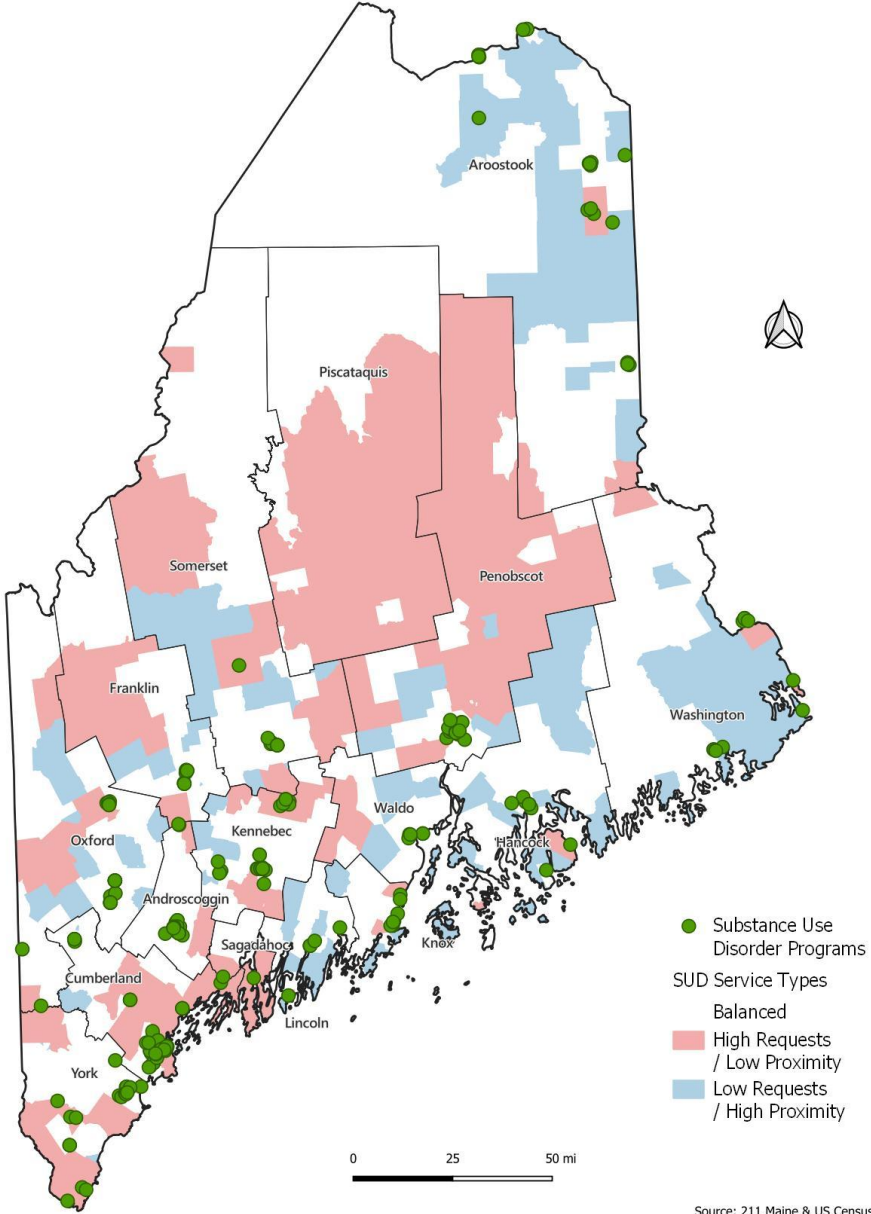


# SUBSTANCE USE DISORDER SERVICES

We were able to identify 256 SUD programs in 211 Maine, 3.5% of the total



Source: 211 Maine & US Census



Source: 211 Maine & US Census

# USING 211 FOR PRIMARY PREVENTION PLANNING IN MAINE

Using 211 request data and U.S. Census community characteristics to better understand ***local drivers*** of 211 requests.

# US CENSUS ACS DATA

how many variables in the American Community Survey?



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About 145,000,000 results (0.52 seconds)

There are **over 18,000** variables in this dataset. Data Profiles contain broad social, economic, housing, and demographic information. The data are presented as population counts and percentages. Mar 17, 2022

<https://www.census.gov> > [developers](#) > [data-sets](#) > [acs-5year](#)

[American Community Survey 5-Year Data \(2009-2020\)](#)

About featured snippets • Feedback

- The US Census has thousands of variables, Maine only has 432 zip codes.
- This makes it very hard for us to identify which of these factors is making people call 211.
- Imagine if you gave 432 people 18,000 different SAT prep classes. How could you tell which one helped people get a better score?

# US CENSUS ACS DATA

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## List of housing characteristics included in the model:

1. Total Vacant Housing Units
2. Housing Built Before 1939
3. Housing Built 2000 or After
4. Renter Occupied Housing Units
5. House Heating Fuel: Wood
6. House Heating Fuel: Oil
7. No Plumbing (Poor Housing Quality)
8. More than 1 Occupant per Room (Overcrowding)
9. Home Value Below \$100,000
10. Home Value Above \$500,000
11. Selected Monthly Owner Costs: More than 30%
12. Gross Rent as a Percentage of Income: More than 30%

Decided to break ACS variable into categories that reflected potential drivers of involvement in primary prevention.

After some exploratory modeling we arrived at 43 variables in 8 categories:

1. Employment
2. Education
3. Demographics
4. Family Composition
5. Health Care Coverage
6. Transportation
7. Housing Characteristics
8. Economic Conditions

# WHAT WE LEARNED

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Significant predictors of 211 calls - Global Model – OLS

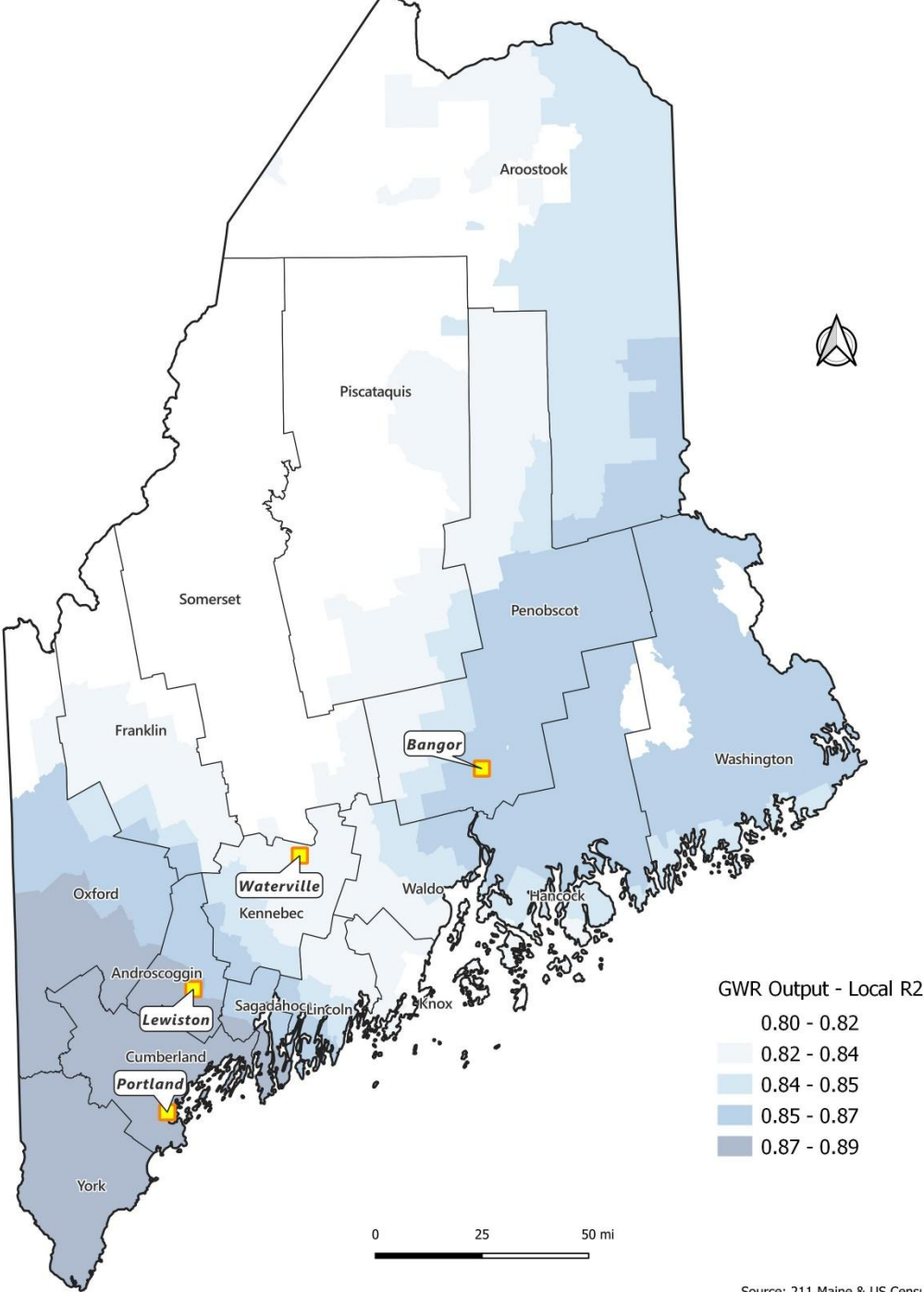
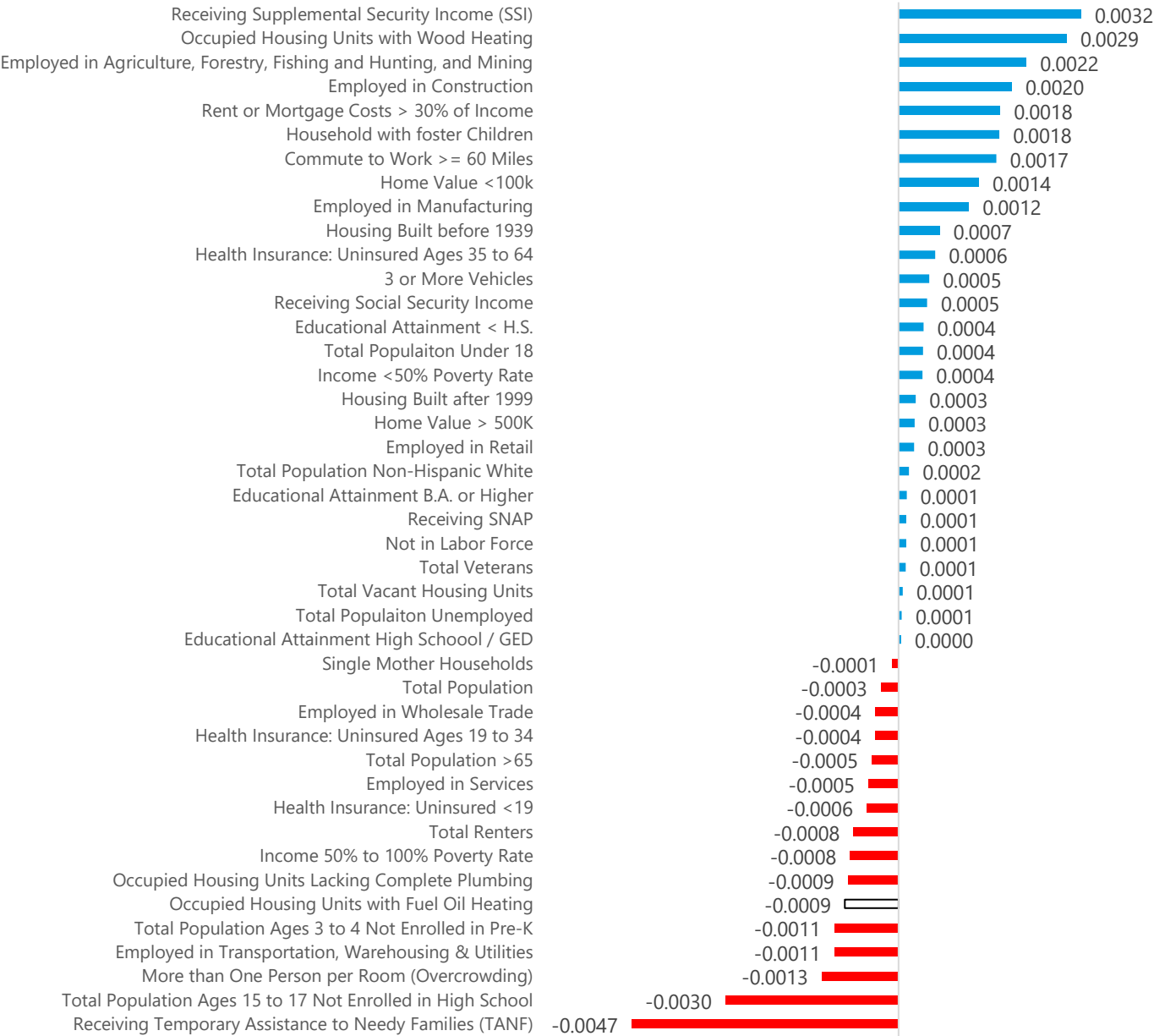
| Variable                                                           | Estimate | Std. Error | t value | P-Value |
|--------------------------------------------------------------------|----------|------------|---------|---------|
| Receiving Supplemental Security Income (SSI)                       | 0.00322  | 0.00130    | 2.48    | 0.01    |
| Occupied Housing Units with Wood Heating                           | 0.00301  | 0.00066    | 4.58    | 0.00    |
| Employed in Agriculture, Forestry, Fishing and Hunting, and Mining | 0.00234  | 0.00114    | 2.05    | 0.04    |
| Employed in Construction                                           | 0.00197  | 0.00098    | 2.02    | 0.04    |
| Rent or Mortgage Costs > 30% of Income                             | 0.00181  | 0.00063    | 2.86    | 0.00    |
| Home Value <100k                                                   | 0.00141  | 0.00044    | 3.25    | 0.00    |
| Employed in Manufacturing                                          | 0.00123  | 0.00060    | 2.07    | 0.04    |
| Housing Built before 1939                                          | 0.00073  | 0.00029    | 2.50    | 0.01    |
| Occupied Housing Units with Fuel Oil Heating                       | -0.00094 | 0.00033    | -2.83   | 0.00    |
| Receiving Temporary Assistance to Needy Families (TANF)            | -0.00468 | 0.00187    | -2.51   | 0.01    |

Our model had an  $R^2$  of 0.76 and returned 10 significant variables.

## Why not just use the Global OLS Model?

- Early models had evidence of autocorrelation.
- Some variables had substantively large effects that were very inconsistently estimated.
- When we divided Maine into pieces and re-ran the same model, coefficient estimates would be statistically different across geographies.

# WHAT WE LEARNED

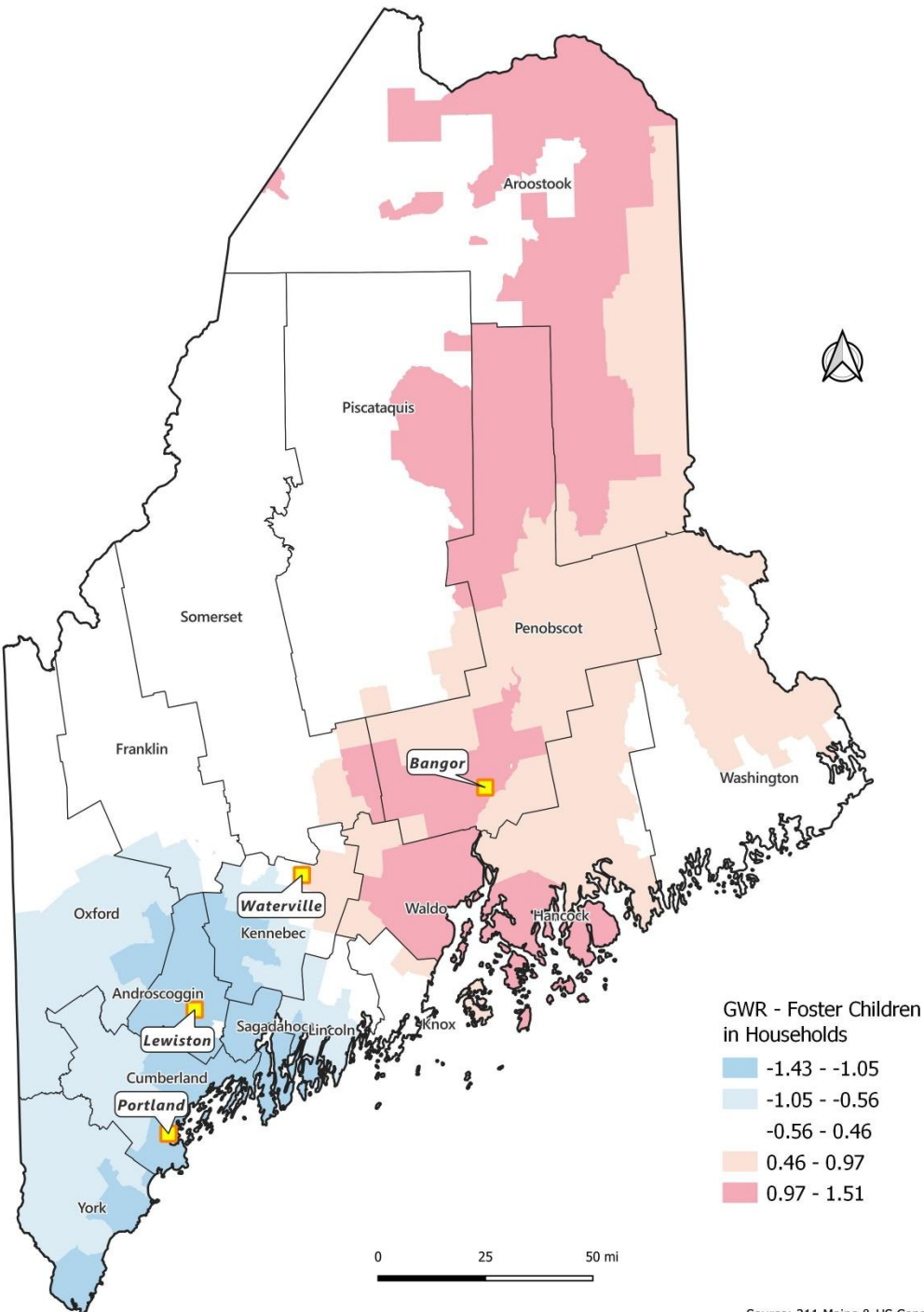
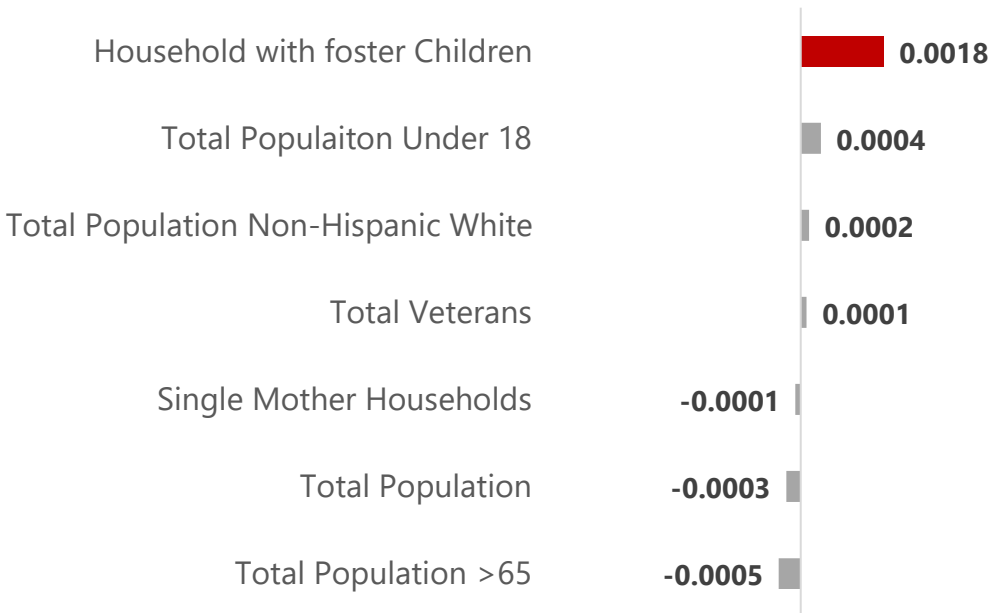


# LOCAL MODEL - GWR

The GWR will allow Maine OCFS to explore which significant variables matter most in predicting 211 calls.

This will support system planning about where to place what type of additional resources or where to have conversations with local service providers to better understand a significant community characteristic.

## Demographics

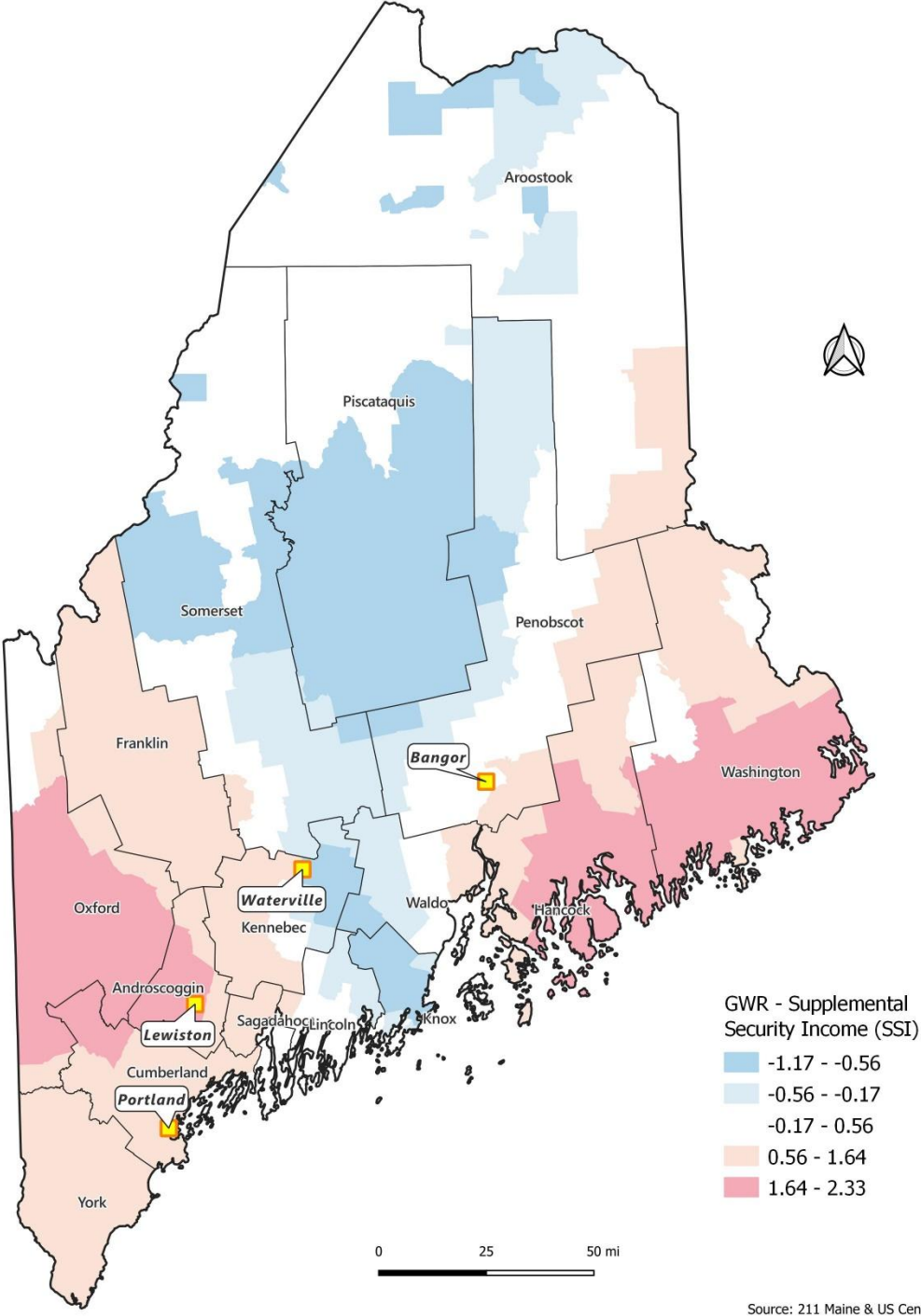
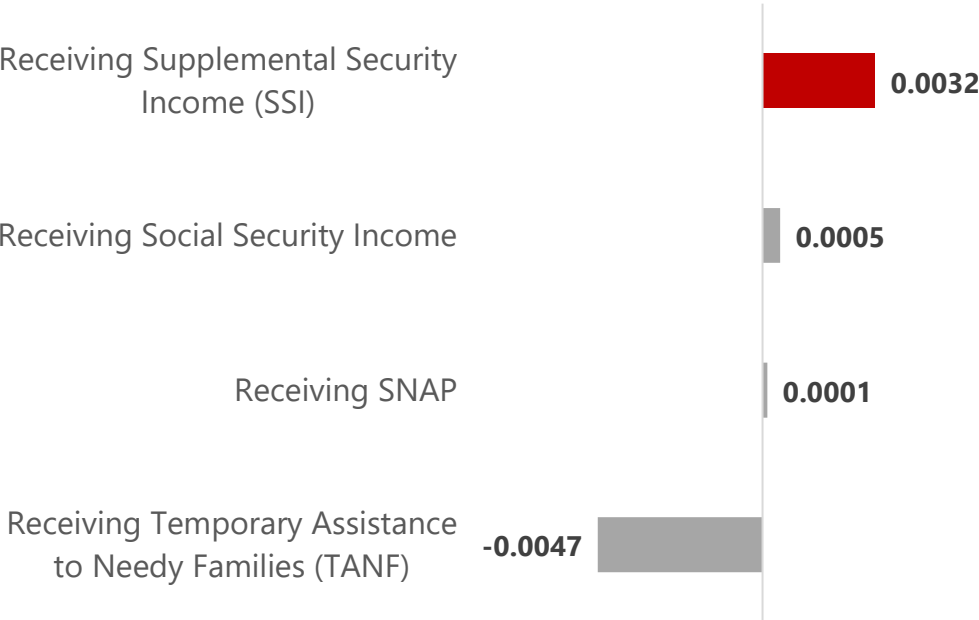


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## Public Assistance Estimates



# IMPLICATIONS

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## **In Maine zip codes that have more...**

- People receiving Supplemental Security Income
- Housing units with Wood Heating (poor quality housing)
- People employed in agriculture, forestry, fishing & hunting, mining and construction (low benefit employment)
- Individuals with rent or mortgage costs in excess of 30% of their income (housing burdened)

## **... also have more 211 requests for services.**

## **In zip codes that have more...**

- People receiving Temporary Assistance to Needy Families (TANF)
- People ages 15 to 17 not enrolled in high school
- Housing units with more than one person per room (overcrowding)
- People employed in Transportation, Warehousing & Utilities
- People ages 3 to 4 not enrolled in pre-school

## **... there are fewer 211 requests for services**

Maine OCFS can use this information to better understand the primary prevention population its trying to reach.

It can also have conversations with people in these groups to better understand why they are, or are not, using 211 to request services.



## 211 DATA: FURTHER OPPORTUNITIES

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- 44 (soon 47) states collect 211 call data.
- The same geographic data is collected and the same call ontology is applied.
- 211counts.org overlays 211 call data with American Community Survey data.
  - That gives a user without data expertise an opportunity to do an abbreviated version of our analyses here.
- 211 was an enthusiastic partner in this research.

# Thank you for your time!

If you have follow-up questions, contact Colin Cepuran ([ccepuran@chapinhall.org](mailto:ccepuran@chapinhall.org)) and Michael Stiehl ([mstiehl@chapinhall.org](mailto:mstiehl@chapinhall.org))